Dynamical Simulation of Neutron Star Hydrostatic Equilibrium via Axiomatic Physical Homeostasis

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Abstract

This paper presents a novel computational model for simulating Neutron Star dynamics using the Axiomatic Physical Homeostasis (APH) framework. Utilizing the Wave-Particle Transport Simulation (WTS) engine, we model the stellar interior as a continuous fluid governed by a geometric "Stiffness" parameter (β) rather than traditional particle-based degeneracy pressure. We demonstrate that this phenomenological approach spontaneously reproduces key astrophysical observables: stable hydrostatic equilibrium, accretion disk formation, polar jet alignment via geometric torsion, and "glitch" events analogous to starquakes. The simulation provides a real-time, numerically stable laboratory for testing the emergent properties of high-density matter under the APH geometric constraints.

1 Introduction

Neutron stars represent the ultimate limit of material density before gravitational collapse into a singularity. Standard models rely on the Tolman-Oppenheimer-Volkoff (TOV) limit derived from General Relativity and Quantum Chromodynamics (QCD). In the Axiomatic Physical Homeostasis (APH) framework, we propose an alternative formulation where the stability of the star is maintained by the *Geometric Stiffness* of the vacuum substrate itself.

We hypothesize that the "degeneracy pressure" preventing collapse is a manifestation of the substrate's resistance to topological compression. This paper details the implementation of this hypothesis in a real-time grid-based simulation and analyzes the resulting phenomenological behaviors.

2 Methodology: The WTS Engine

The simulation utilizes the Wave-Particle Transport Simulation (WTS) engine, implemented in Taichi/Python. The core physics loop solves a modified Navier-Stokes equation where pressure is derived from a programmable Equation of State (EOS).

2.1 The APH Hydrodynamic Equation

The evolution of the density field ρ is governed by the interplay of Gravity, Geometric Stiffness, and Torsion. The update rule for velocity $\mathbf{v}$ at grid point (i, j) is given by:

$$\frac{\partial \mathbf{v}}{\partial t} = \mathbf{F}_{gravity} + \mathbf{F}_{pressure} + \mathbf{F}_{torsion} - \gamma \mathbf{v} \tag{1}$$

Where γ is a damping factor (≈ 0.99) representing energy dissipation via neutrino cooling or gravitational waves.

2.2 Gravity and the Geometric Stiffness

Gravity is modeled as a standard central potential scaling with the local density:

$$\mathbf{F}_{gravity} = -\frac{G \cdot \rho}{r + \epsilon} \hat{r} \tag{2}$$

Crucially, the resistive force ($\mathbf{F}_{pressure}$) is not linear. It is governed by a local stiffness parameter β, which scales non-linearly with density to model the Pauli Exclusion Principle:

$$\beta(\rho) = \beta_0 + k\rho^{\Gamma} \tag{3}$$

In our calibrated simulation, we utilized a polytropic index $\Gamma \approx 2.4$, consistent with stiff nuclear matter.

The restoration force is calculated via the Laplacian of the density field, saturated by a hyperbolic tangent function to enforce the APH stability constraint (preventing singularities):

$$\mathbf{F}_{pressure} = \beta \cdot \tanh(\lambda \nabla^2 \rho) \tag{4}$$

This term acts as a "Geometric Spring," pushing back against gravitational compression.

3 Simulation Results

3.1 Hydrostatic Equilibrium

Upon initialization with a dense seed core, the simulation rapidly converges to a stable state. The radial density profile (visualized via real-time telemetry) exhibits a sharp "cliff" at the stellar surface, dropping from $\rho_{core} \approx 60.0$ to $\rho_{vac} \approx 0.0$ over a few grid cells. This matches the physical expectation of a neutron star "crust," distinct from the diffuse atmospheres of main-sequence stars.

3.2 Accretion and the Disk

We introduced a continuous matter injection term at the simulation boundary:

$$\frac{\partial \rho}{\partial t} \leftarrow \rho + \alpha \sin(\omega t + \phi) \tag{5}$$

This matter spirals inward due to the conservation of angular momentum imposed by the grid logic. The simulation spontaneously forms a stable, rotating accretion disk (visualized in purple/red) that feeds the core without destabilizing the equilibrium.

3.3 Geometric Torsion and Jets

The simulation introduces a "Torsion" term τ proportional to the spin rate. We observed that this torsional field creates a channel of low resistance along the axis of rotation. When visualized, this manifests as collimated polar jets (Cherenkov radiation), effectively replicating the pulsar mechanism via geometric frame-dragging.

3.4 Starquakes (The Geometric Bounce)

By injecting a localized density spike ("trigger_starquake"), we tested the system's resilience. **Observation:** The core momentarily compresses, violating the stiffness limit. **Response:** The WTS engine generates a repulsive shockwave that propagates to the crust, causing the star to oscillate (ring) before settling back to a slightly denser equilibrium. **Implication:** This validates the APH hypothesis that starquakes and planetary formation events (e.g., the Moon bounce) are topologically identical "Geometric Bounces" driven by the restoration of homeostasis.

4 Discussion: APH Telemetry

The simulation tracks three key APH metrics in real-time: **Core Density (ρ):** Proxy for total mass and gravitational potential. **Stiffness (β):** Proxy for the quantum degeneracy pressure. The simulation confirms that β must scale dynamically with ρ to prevent collapse. **Associator Hazard (A):** Defined as the local divergence between gravity and pressure ($\nabla \cdot \mathbf{F}_{net}$). High hazard values were observed at the "crust," identifying it as the region of maximal topological stress.

5 Conclusion

The "APH Neutron Star" simulation demonstrates that complex astrophysical phenomena can be accurately modeled as emergent properties of a geometric substrate. By replacing particle-wise QCD calculations with a macroscopic "Stiffness" field, we achieved a real-time, numerically stable model of a pulsar that exhibits hydrostatic equilibrium, accretion dynamics, and jet formation. This strongly supports the utility of the WTS engine as a tool for *Computational Astrophysics via Geometry*.